Event-Driven and Many-Threaded Architectures for High-Frequency Trading

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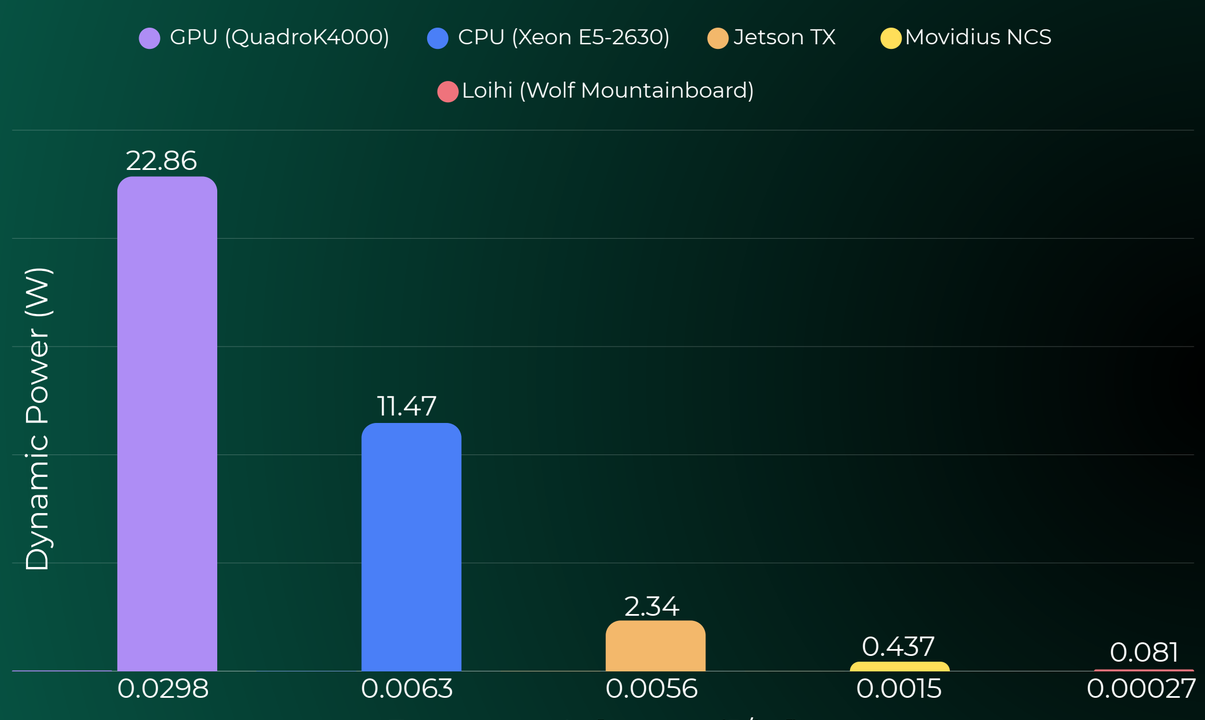
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BACKGROUND

High-Frequency Trading (HFT) is a type of automated (algorithmic) trading that exploits short-lived pricing inefficiencies to generate small, accumulative income over time. Today, firms can execute trades at nanosecond-scale tick-to-trade latencies using **Field-Programmable Gate Arrays (FPGAs)** strategically placed inside stock exchanges' co-location facilities, physically constrained by fixed rack space and power-draw limits. Increasing compute density compute density within that same space would enable the implementation of more complex, multi-symbol correlation strategies.

Event-driven architectures, unlike FPGAs, are asynchronous; computation fires on stimulus, and idle regions consume very little power. Replacing clocked execution with 12,288 event-driven threads on FPGA softcores achieved an order-of-magnitude energy reduction over a Xeon cluster at matched power draw [1]. On fabricated 14nm silicon, spike-based computation achieves over three orders of magnitude better energy-delay product than conventional CPUs at matched process node and voltage [2]. Real-time simulation of millions of spiking neurons has been sustained at approximately 1W per 18-core chip [3].

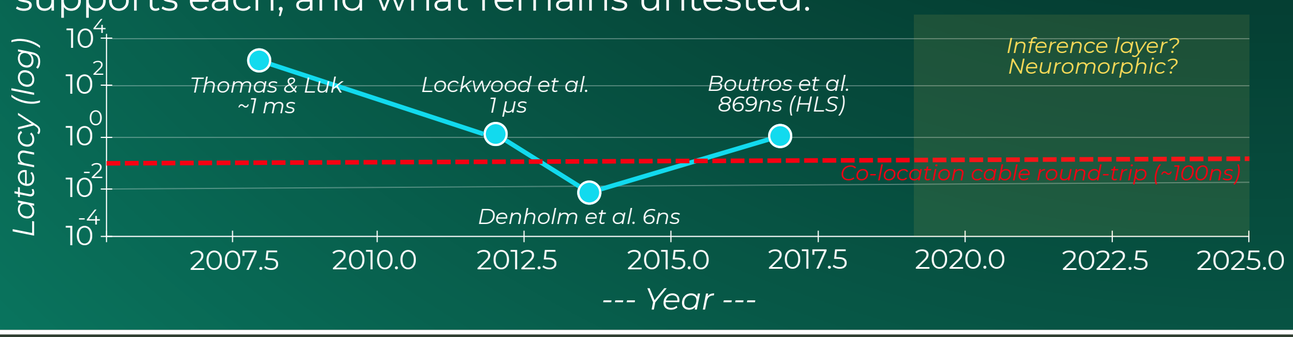


"Mean power consumption and energy cost per inference across hardware devices." [4], dynamic power measurements sit atop the bars, energy per inference sit below the x-axis.

RESEARCH AIM

This survey aims to synthesise the HFT domain with event-based and neuromorphic architectures. Within a fixed co-location power budget, can a functionally decomposed pipeline (FPGA for deterministic execution, neuromorphic hardware for inference) outperform a pure-FGPA system that cannot add pattern-recognition logic without breaking timing closure?

Synthesising nineteen papers across FPGA acceleration, event-driven computing, and neuromorphic architecture, we evaluate whether the components for such a system exist, what evidence supports each, and what remains untested.



KEY FINDINGS

The research corpus spans three fields that rarely cite each other. Most of the literature optimises one component of the trading pipeline in isolation.

The key findings are results that survived cross-domain scrutiny where, measured hardware performance, financial backtesting evidence, market-stability analysis, and the likes converge on the question:

What happens when the fastest execution hardware and the most power-efficient inference hardware coexist in the same market?

1. **FPGA datapaths have reduced feed-processing latency from milliseconds to 6 ns ...**
   ... on real NASDAQ data; they are approaching signal propagation limits in co-location cabling. High-Level Synthesis compares with latency to hand-coded RTL at 869ns round-trip, using less than 8% of FPGA logic [4].

2. **Event-driven execution scales sub-linearly across four case studies on POETS ...**
   ... whereas synchronous approaches scale super-linearly on those cases. On the same hardware, 509,648 spiking neurons were simulated in real-time at 83.8% of theoretical capacity [5].

3. **Unsupervised STDP-trained SNNs returned 76.8% in 19-day simulated trading ...**
   ... on AAPL tick data, outperforming supervised variants at 42.5% [6]. Sharpe ratios above 20 were reported on crude oil and gold futures with the same learning rule.

4. **Algorithmic trading over 65% of market volume triggers a flash-crash probability of 76.2% ...**
   ... on five years of Chinese A-share data [7]. The 2010 Flash Crash was traced to a single algorithm selling $4.1 billion of E-mini contracts; high-frequency traders amplified but did not originate it.

RESEARCH GAPS

- No cross-substrate power benchmark measures the same financial workload across FPGA, neuromorphic ASIC, and GPU with real power draw. Efficiency comparisons in this corpus use different tasks, metrics, or simulated numbers.
- **(Actionable Now)** Event-driven financial simulation never run on POETS
- No fabricated neuromorphic chip tested on a financial workload
- **(Actionable Now)** SNN trading never validated using real transaction costs or slippage over a significant amount of time and across a variety of equities
- **No study** models the effects of hardware-heterogeneous trading populations on the market

CONCLUSION

Undoubtedly, purpose built hardware accelerates financial workloads by orders of magnitude. FPGA datapaths reach nanosecond-scale deterministic latency on market data, and event-driven architectures scale sub-linearly where synchronous systems cannot. SNNs detect temporal patterns in tick data that supervised models miss. Each one of these results is measured and is absolute, yet none has been combined into a single system.

The path forward is functional decomposition. A proposition: FPGAs handle deterministic execution paths for minimum jitter, neuromorphic processors power-efficiently handle inference, and event-driven messaging connects the two within the co-location power budgets. The next step is to build and measure that split pipeline on a standardised financial task, then ask whether the market it trades can absorb it.

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